

## SAFETY DATA SHEET

Creation Date 21-Jan-2009

Revision Date 20-Jan-2025

Revision Number 1

### 1. Identification

**Product Name** 1-Butanol, GR

**Cat No. :** W00403

**CAS No** 71-36-3  
**Synonyms** n-Butanol; n-Butyl alcohol, Butan-1-ol

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                                   |            |
|-------------------------------------------------------------------|------------|
| Flammable liquids                                                 | Category 3 |
| Acute oral toxicity                                               | Category 4 |
| Skin Corrosion/Irritation                                         | Category 2 |
| Serious Eye Damage/Eye Irritation                                 | Category 1 |
| Specific target organ toxicity (single exposure)                  | Category 3 |
| Target Organs - Respiratory system, Central nervous system (CNS). |            |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Flammable liquid and vapor  
Harmful if swallowed

Causes skin irritation  
Causes serious eye damage  
May cause respiratory irritation  
May cause drowsiness or dizziness

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Wear protective gloves/protective clothing/eye protection/face protection  
Do not breathe dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge  
Keep cool

**Response**

Get medical attention/advice if you feel unwell

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
Call a POISON CENTER or doctor/physician if you feel unwell

**Skin**

If skin irritation occurs: Get medical advice/attention  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower  
Wash contaminated clothing before reuse

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER or doctor/physician

**Ingestion**

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell  
Rinse mouth

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/information on Ingredients

| Component       | CAS No  | Weight % |
|-----------------|---------|----------|
| n-Butyl alcohol | 71-36-3 | 99       |

#### 4. First-aid measures

|                                            |                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                      |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting      |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                             |

#### 5. Fire-fighting measures

|                                         |                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                              |
| <b>Flash Point</b>                      | 35 °C / 95 °F                                                                         |
| <b>Method -</b>                         | CC (closed cup)                                                                       |
| <b>Autoignition Temperature</b>         | 340 °C / 644 °F                                                                       |
| <b>Explosion Limits</b>                 |                                                                                       |
| <b>Upper</b>                            | 11.2 vol %                                                                            |
| <b>Lower</b>                            | 1.4 vol %                                                                             |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                              |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                              |

#### Specific Hazards Arising from the Chemical

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

#### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

**Health**  
2

**Flammability**  
3

**Instability**  
0

**Physical hazards**  
N/A

#### 6. Accidental release measures

|                                             |                                                                                          |
|---------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation.              |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                             |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. |

## 7. Handling and Storage

**Handling** Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Flammables area. Incompatible Materials. Strong oxidizing agents. Reducing Agent. Acid chlorides. copper. Copper alloys. Acid anhydrides.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component       | ACGIH TLV   | OSHA PEL                                                                                                                    | NIOSH                                                               | Mexico OEL (TWA) |
|-----------------|-------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------------|
| n-Butyl alcohol | TWA: 20 ppm | Skin<br>(Vacated) Ceiling: 50 ppm<br>(Vacated) Ceiling: 150 mg/m <sup>3</sup><br>TWA: 100 ppm<br>TWA: 300 mg/m <sup>3</sup> | IDLH: 1400 ppm<br>Ceiling: 50 ppm<br>Ceiling: 150 mg/m <sup>3</sup> | TWA: 20 ppm      |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures** Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location.

### Personal Protective Equipment

**Eye/face Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection** Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:** Organic gases and vapours filter. Type A. Brown. conforming to EN14387.

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                          |                          |
|--------------------------|--------------------------|
| Physical State           | Liquid                   |
| Appearance               | Colorless                |
| Odor                     | Alcohol-like             |
| Odor Threshold           | No information available |
| pH                       | No information available |
| Melting Point/Range      | -89 °C / -128.2 °F       |
| Boiling Point/Range      | 117.6 °C / 243.7 °F      |
| Flash Point              | 35 °C / 95 °F            |
| Method -                 | CC (closed cup)          |
| Evaporation Rate         | 0.46                     |
| Flammability (solid,gas) | Not applicable           |

**Flammability or explosive limits**

|                                        |                           |
|----------------------------------------|---------------------------|
| Upper                                  | 11.2 vol %                |
| Lower                                  | 1.4 vol %                 |
| Vapor Pressure                         | 6.7 mbar @ 20 °C          |
| Vapor Density                          | 2.6                       |
| Specific Gravity                       | 0.810                     |
| Solubility                             | Slightly soluble in water |
| Partition coefficient; n-octanol/water | No data available         |
| Autoignition Temperature               | 340 °C / 644 °F           |
| Decomposition Temperature              | No information available  |
| Viscosity                              | 2.95 mPa.s (20 °C)        |
| Molecular Formula                      | C4 H10 O                  |
| Molecular Weight                       | 74.12                     |
| Refractive index                       | 1.390 - 1.400             |

## 10. Stability and reactivity

|                                         |                                                                                                 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                      |
| <b>Stability</b>                        | Stable under normal conditions.                                                                 |
| <b>Conditions to Avoid</b>              | Keep away from open flames, hot surfaces and sources of ignition. Incompatible products.        |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Reducing Agent, Acid chlorides, copper, Copper alloys, Acid anhydrides |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                         |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                        |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                   |

## 11. Toxicological information

**Acute Toxicity****Product Information****Component Information**

| Component       | LD50 Oral                | LD50 Dermal                  | LC50 Inhalation             |
|-----------------|--------------------------|------------------------------|-----------------------------|
| n-Butyl alcohol | LD50 = 700 mg/kg ( Rat ) | LD50 = 3402 mg/kg ( Rabbit ) | LC50 > 8000 ppm ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | Severe eye irritant; Irritating to respiratory system and skin                           |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component       | CAS No  | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------|---------|------------|------------|------------|------------|------------|
| n-Butyl alcohol | 71-36-3 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system Central nervous system (CNS)  
**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not flush into surface water or sanitary sewer system.

| Component       | Freshwater Algae                                                                                                                                                                        | Freshwater Fish                                                                                                                                                                                                                                                                                                 | Microtox                                                                                              | Water Flea                                                                                                                                                    |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| n-Butyl alcohol | EC50: 225 mg/L, 96h<br>(Pseudokirchneriella subcapitata) OECD Guideline 201<br>EC50: > 500 mg/L, 72h<br>(Desmodesmus subspicatus)<br>EC50: > 500 mg/L, 96h<br>(Desmodesmus subspicatus) | LC50: 1376 mg/L, 96h<br>(Pimephales promelas) OECD Guideline 203 :<br>100000 - 500000 µg/L, 96h static (Lepomis macrochirus)<br>LC50: = 1740 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: = 1910000 µg/L, 96h static (Pimephales promelas)<br>LC50: 1730 - 1910 mg/L, 96h static (Pimephales promelas) | EC50 = 2041.4 mg/L 5 min<br>EC50 = 2186 mg/L 30 min<br>EC50 = 3980 mg/L 24 h<br>EC50 = 4400 mg/L 17 h | EC50: 1328 mg/L, 48h<br>(Daphnia magna) OECD Guideline 202<br>EC50: 1897 - 2072 mg/L, 48h Static (Daphnia magna)<br>EC50: = 1983 mg/L, 48h<br>(Daphnia magna) |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Will likely be mobile in the environment due to its water solubility.

| Component       | log Pow |
|-----------------|---------|
| n-Butyl alcohol | 1       |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component                 | RCRA - U Series Wastes | RCRA - P Series Wastes |
|---------------------------|------------------------|------------------------|
| n-Butyl alcohol - 71-36-3 | U031                   | -                      |

## 14. Transport information

### DOT

UN-No UN1120  
 Proper Shipping Name BUTANOLS  
 Hazard Class 3  
 Packing Group III

### TDG

UN-No UN1120  
 Proper Shipping Name BUTANOLS  
 Hazard Class 3  
 Packing Group III

**IATA**

UN-No UN1120  
 Proper Shipping Name BUTANOLS  
 Hazard Class 3  
 Packing Group III

**IMDG/IMO**

UN-No UN1120  
 Proper Shipping Name BUTANOLS  
 Hazard Class 3  
 Packing Group III

**15. Regulatory Information****United States of America Inventory**

| Component       | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|-----------------|---------|------|-----------------------------------------------|-----------------------------|
| n-Butyl alcohol | 71-36-3 | X    | ACTIVE                                        | -                           |

**Legend:**

TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'- - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

TSCA 12(b) - Notices of Export

Not applicable

**International Inventories**

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component       | CAS No  | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|-----------------|---------|-----|------|-----------|-------|------|------|------|-------|----------|
| n-Butyl alcohol | 71-36-3 | X   | -    | 200-751-6 | X     | X    | X    | X    | X     | KE-03867 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component       | CAS No  | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|-----------------|---------|----------|-------------------------------|---------------------------------|
| n-Butyl alcohol | 71-36-3 | 99       | 1.0 %                         | -                               |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA - Occupational Safety and Health Administration** Not applicable

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355).

| Component       | Hazardous Substances RQs | CERCLA Extremely Hazardous Substances RQs | SARA Reportable Quantity (RQ) |
|-----------------|--------------------------|-------------------------------------------|-------------------------------|
| n-Butyl alcohol | 5000 lb                  | -                                         | 5000 lb<br>2270 kg            |

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component       | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-----------------|---------------|------------|--------------|----------|--------------|
| n-Butyl alcohol | X             | X          | X            | -        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): Y

DOT Marine Pollutant N

DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

**Other International Regulations****Mexico - Grade**

Serious risk, Grade 3

**Authorisation/Restrictions according to EU REACH**

| Component       | CAS No  | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------|---------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| n-Butyl alcohol | 71-36-3 | -                                                                   | Use restricted. See entry 75.<br>(see link for restriction details)           | -                                                                                                     |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component       | CAS No  | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------|---------|----------|------------------------------|---------------------------|--------------------------------------------|
| n-Butyl alcohol | 71-36-3 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component | CAS No | Seveso III Directive | Seveso III Directive | Rotterdam | Basel Convention |
|-----------|--------|----------------------|----------------------|-----------|------------------|
|-----------|--------|----------------------|----------------------|-----------|------------------|



|                 |         | (2012/18/EC) -<br>Qualifying Quantities<br>for Major Accident<br>Notification | (2012/18/EC) -<br>Qualifying Quantities<br>for Safety Report<br>Requirements | Convention (PIC) | (Hazardous Waste) |
|-----------------|---------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------|-------------------|
| n-Butyl alcohol | 71-36-3 | Not applicable                                                                | Not applicable                                                               | Not applicable   | Not applicable    |

## 16. Other Information

**Prepared By** Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Creation Date** 21-Jan-2009  
**Revision Date** 20-Jan-2025  
**Print Date** 20-Jan-2025  
**Revision Summary** Initial Release.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**